

Why Your RAG Agent Is Confidently Wrong

Retrieval Choices That Actually Matter

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David vonThenen

- Are you Human or an AI?
- I want 5 Kubernetes
- Virtual Machines are Real
- Cloudy, cloudy, cloudy...
- There is storage for that!

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Agenda

- **Semantically Similar Aren't Answers**
- **Answers Need Contextual Understanding**
 - **Demo:** The Similarity Fallacy
 - **Demo:** Querying Structured Data
- **Solution: Hybrid Retrieval**
 - **Demo:** Hybrid Retrieval
- **Q&A**

Semantically Similar Isn't The Same As Answers

What is Semantic Similarity?

- Objects/Vectors That Have:
 - Similar Numerical Representation
 - Captures Relationships Between:
 - Words, Concepts, etc
- Is The Foundation of Vector Search
 - Query Nearest Vectors (kNN), like:
 - Cosine Similarity
 - Euclidean Distance
- How Does This Relate To Answers?

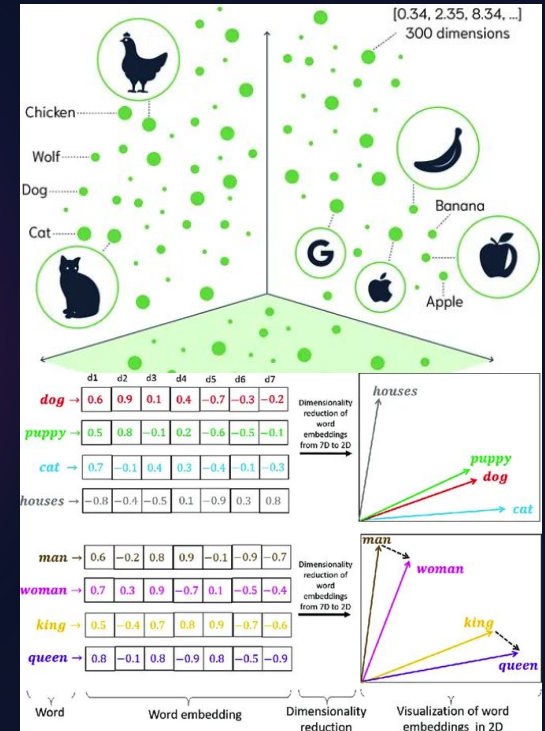


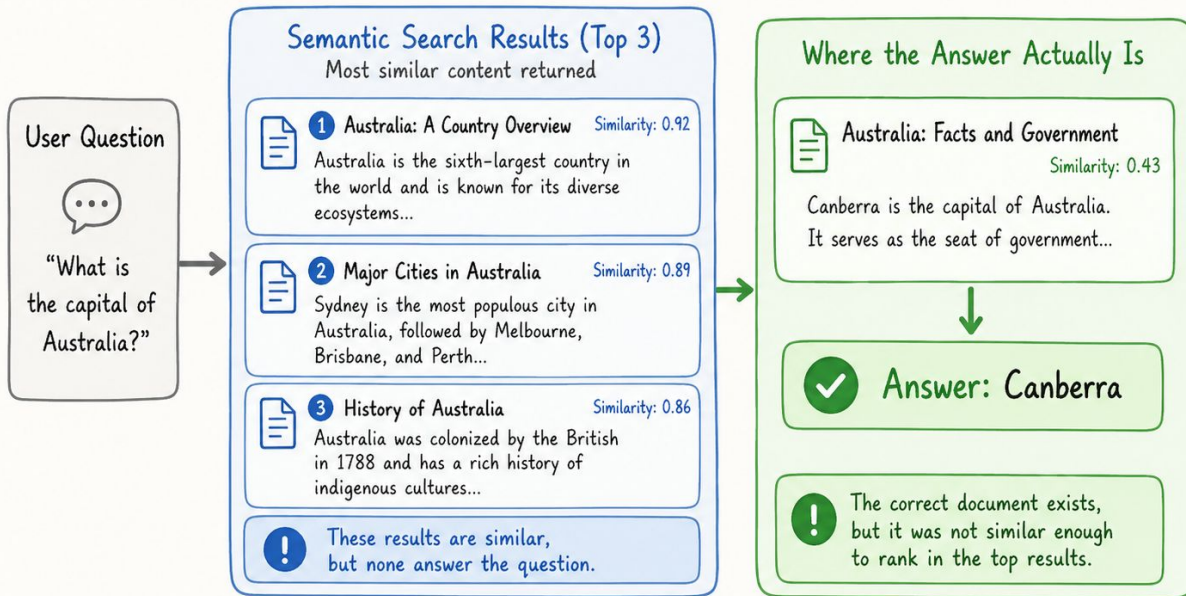
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<https://mehmetozkaya.medium.com/understanding-semantic-meaning-and-similarity-search-cosine-similarity-and-euclidean-distance-666fcb5911ea>
<https://fastdatascience.com/natural-language-processing/semantic-similarity-with-sentence-embeddings/>

The Fallacy...

Semantic Search: Similar Content $\neq$ The Answer

Vector similarity can retrieve relevant-looking passages without returning the actual answer.



Failure Case:

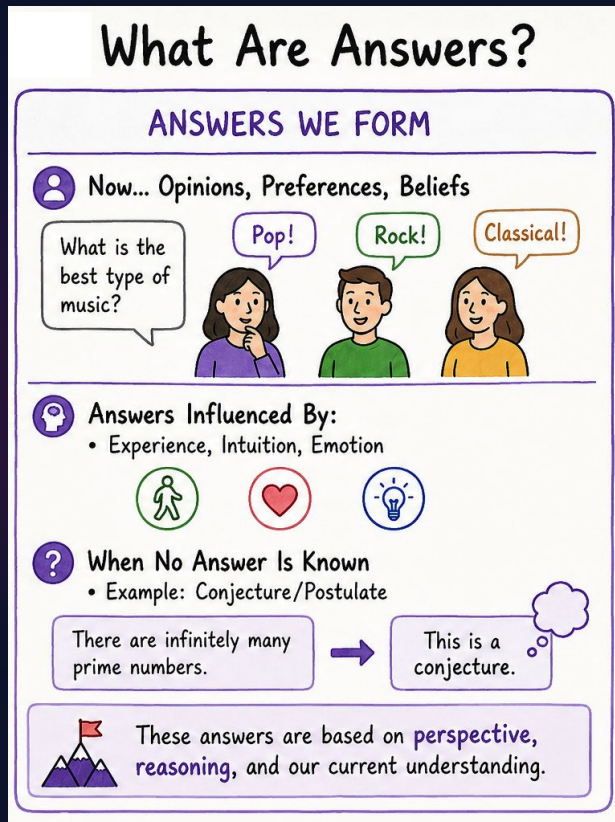
Semantic search retrieved similar topics, not the answer.

Answers Need Contextual Understanding

Context Is Important!

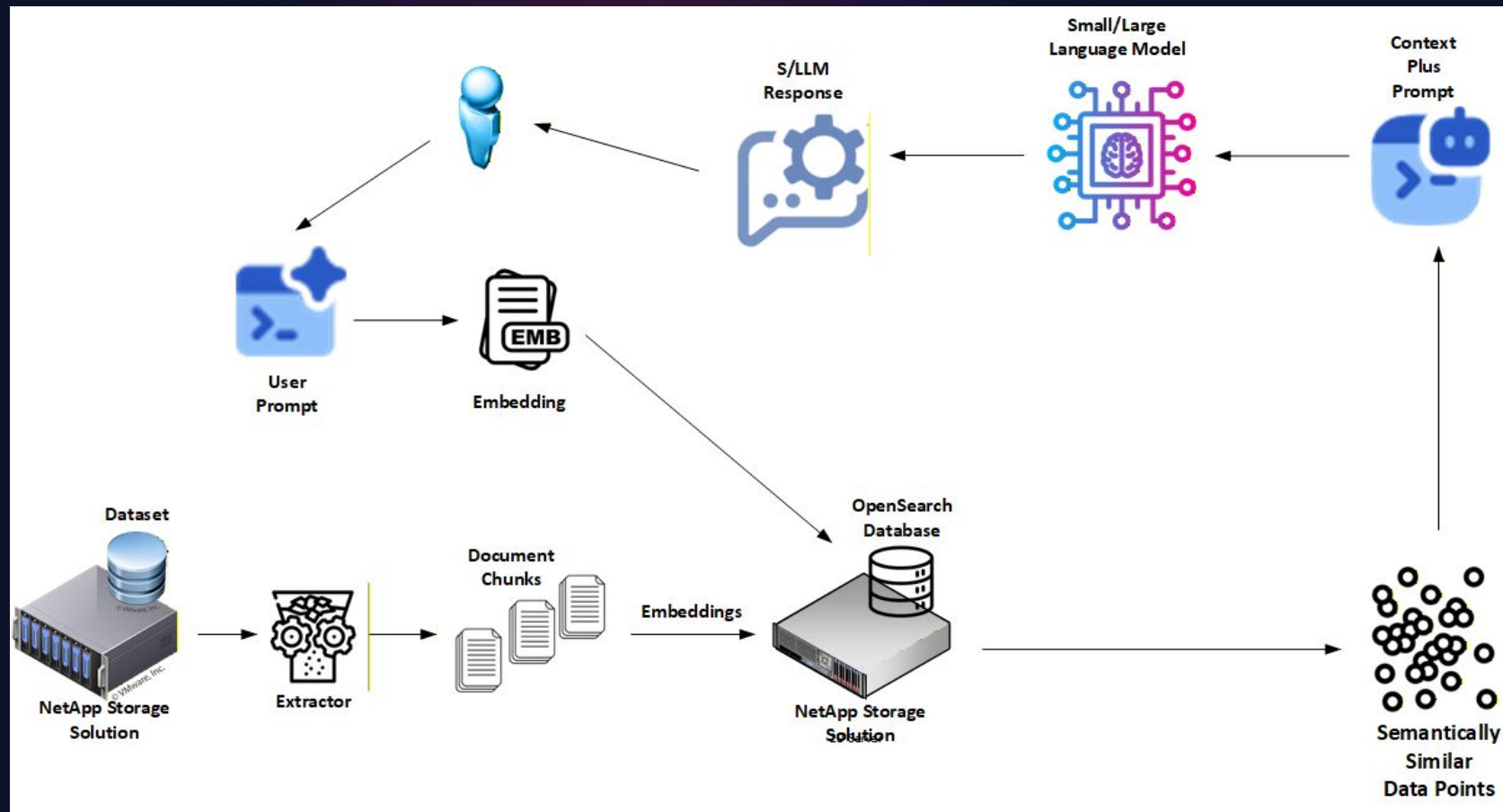
- Conceptually, What Are Answers?
 - Factual Replies To Questions
 - Solution To A Problem
 - Example: Math Problem
- Now... Opinions, Preferences, Beliefs
 - Answers Influenced By:
 - Experience, Intuition, Emotion
 - When No Answer Is Known
 - Example: Conjecture/Postulate

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Vector RAG Architecture

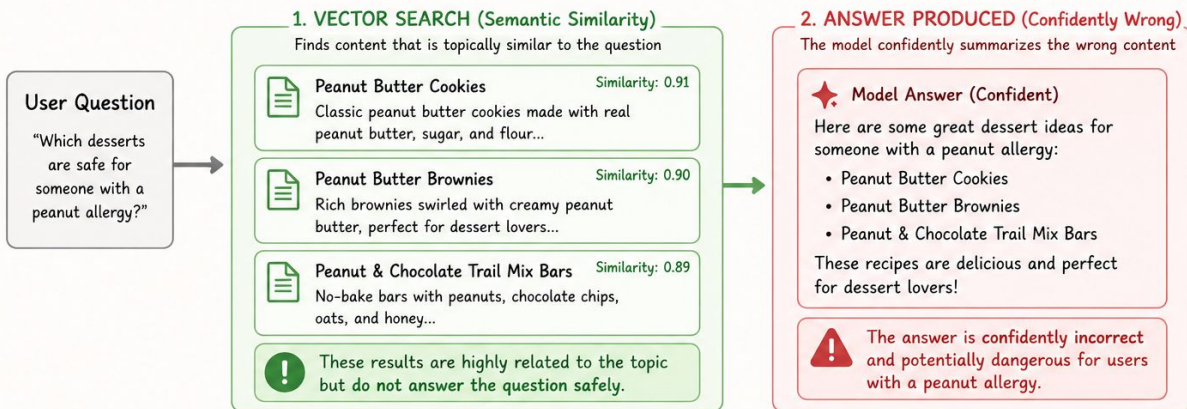
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Similar Doesn't Mean Facts!

Vector-Only Search Failure: Confidently Incorrect Answers

Semantic similarity finds related content, not the right answer.



WHY VECTOR-ONLY SEARCH FAILS



What Semantic Similarity Sees

- "Peanut allergy" matches strongly with pages about peanuts and peanut-based recipes.
- Topical relevance $\neq$ Safe for peanut allergy.



Where the True Answer Lives

- Allergen fields or ingredient lists that explicitly state "contains_peanuts: false" and "contains_tree_nuts: false"
- Safe dessert recipes with nut-free labels or notes.



Vector search finds what is *about* the topic, not what is *safe or correct*.



Use Hybrid RAG: Use keyword/structured filters on allergens and ingredients to find safe recipes, then use vector search to add helpful context like taste, substitutes, and cooking tips.

Demo: The Similarity Fallacy

<https://youtu.be/h1MasHKu-Xo>

Graph Retrieval is a Certainty!

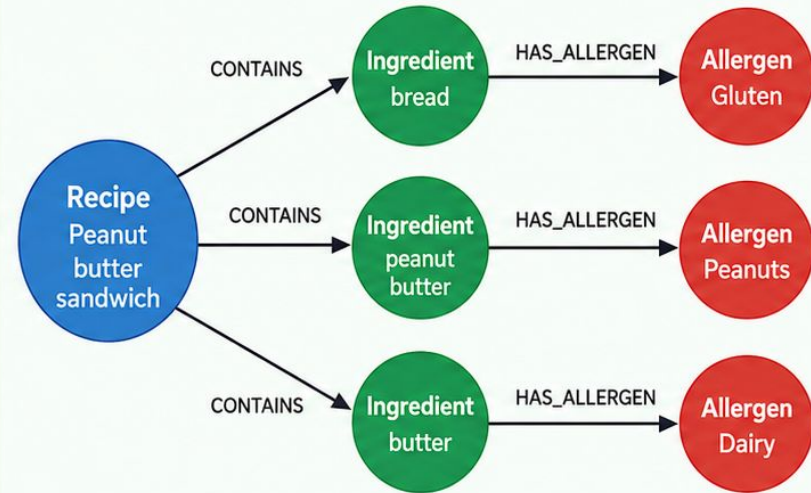
Raw text

Recipe: Peanut butter sandwich
Ingredients: bread, peanut butter, butter
Allergens: gluten, peanuts, dairy

Unstructured text mixes recipe, ingredients, and allergens together.

Extract
structure

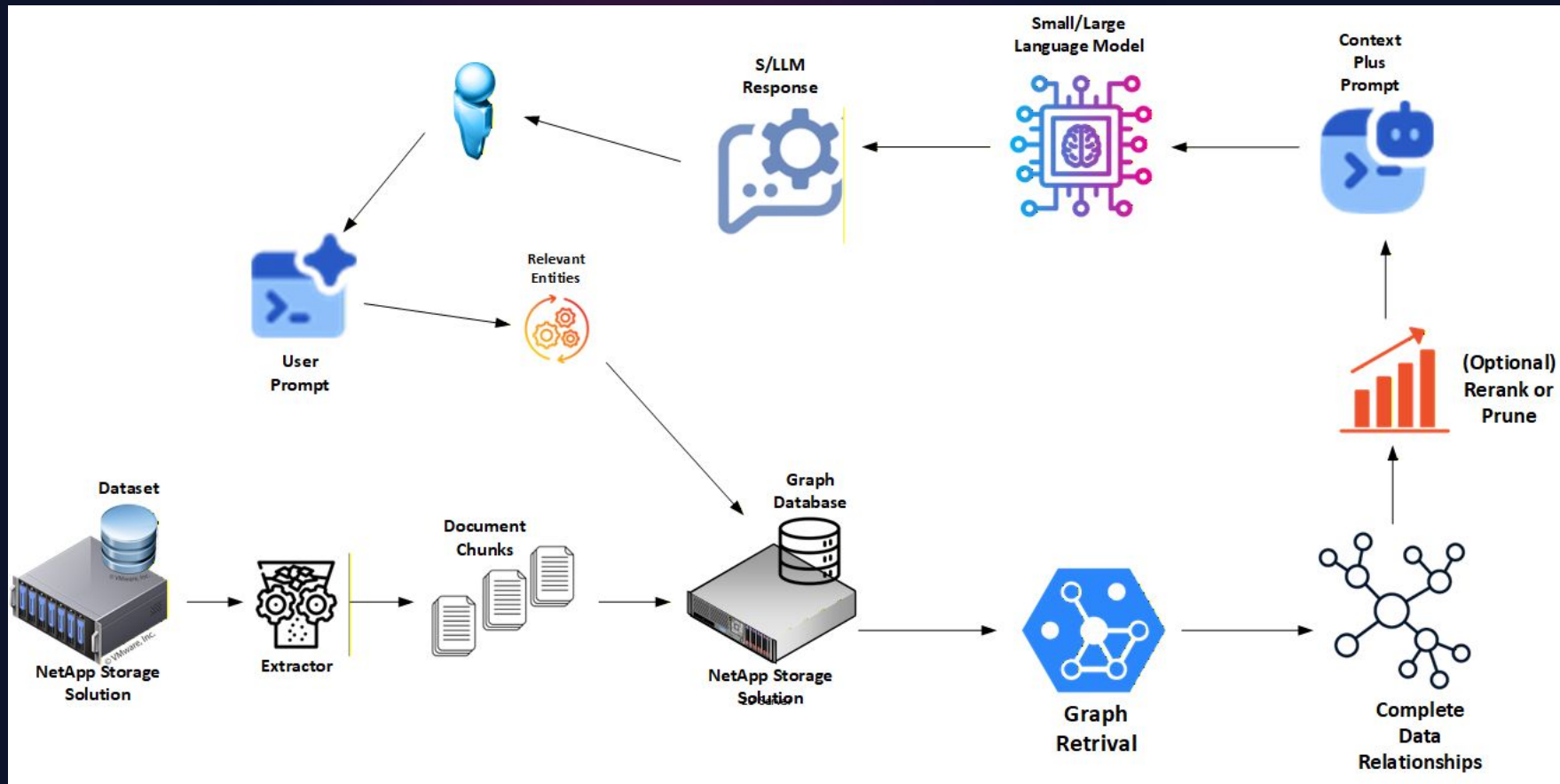
Graph relationships



Each concept from the raw text becomes an explicit relationship in the graph.

Graph RAG Architecture

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Graph-based Retrieval

- What Are Knowledge Graphs?
 - Retrieval Based on Data Relationships
 - Association Between Data Points
 - Retrieve a Network of Facts
- Benefits:
 - Highest Retrieval Relevance
 - Most Accurate Collection of Facts
 - Best Answers for Users
- Drawbacks:
 - Requires Structured Data
 - Manage Ontologies (Like, in SQL)

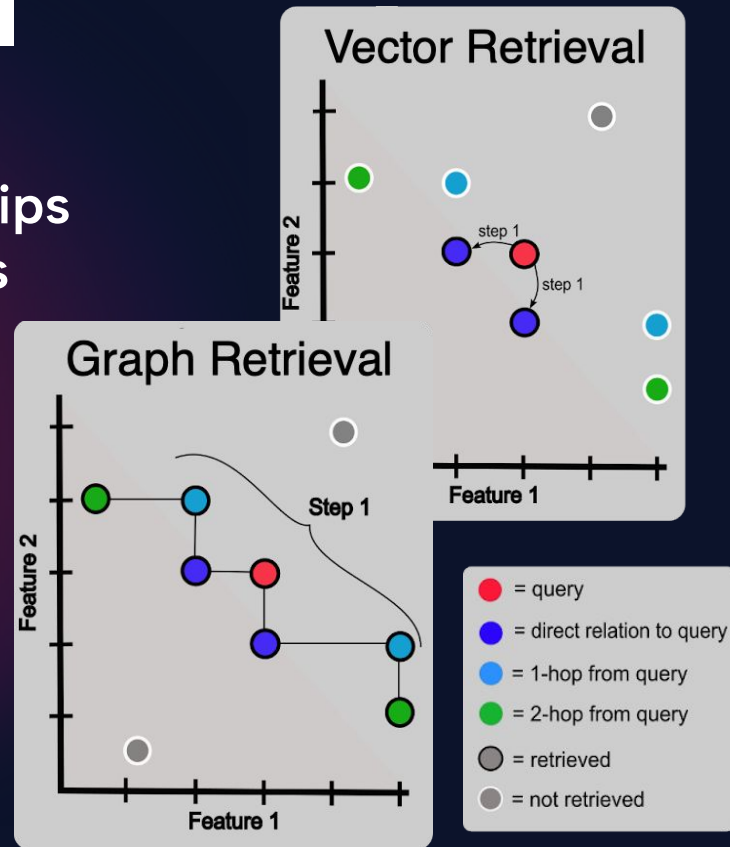


Image Credit:

[Nick Alonso](#), [Beren Millidge](#)

[Understanding Graph-based RAG and Multi-Hop Question Answering](#)

Demo: Querying Structured Data

<https://youtu.be/GWnJCAwmH80>

Solution: Hybrid Retrieval

Hybrid (Graph+Vector) RAG

Image Attribution
HybridRAG: Integrating Knowledge
Graphs and Vector Retrieval
Augmented Generation for
Efficient Information Extraction
<https://arxiv.org/html/2408.04948v1>

BETA



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arXiv:2408.04948v1 [cs.CL] 09 Aug 2024

HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction

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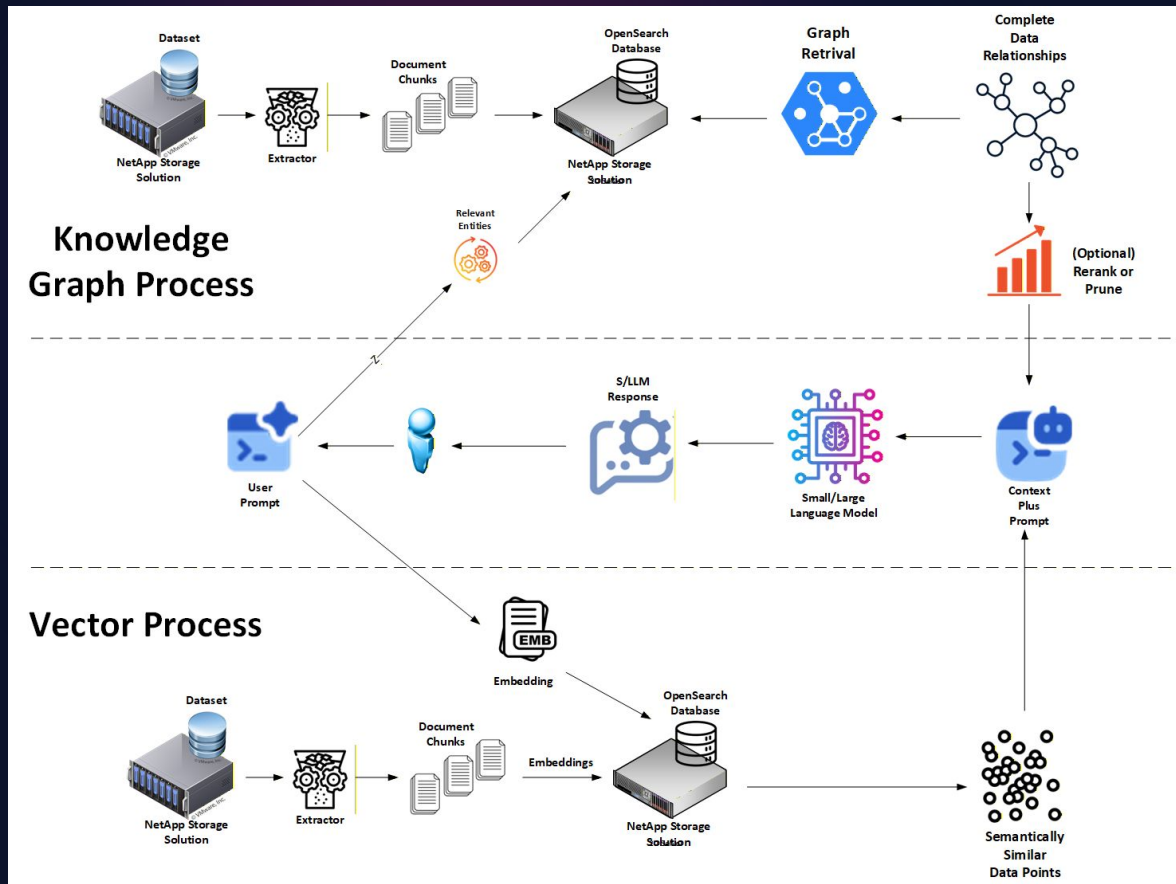
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Hybrid RAG Architecture

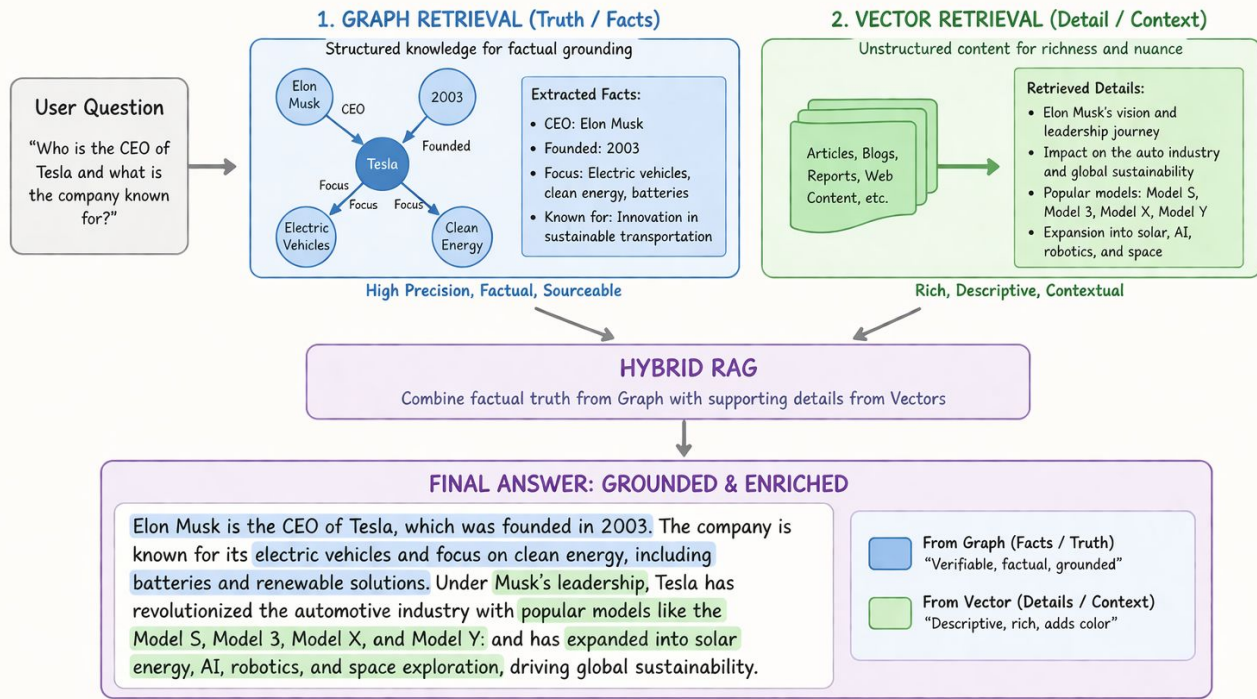
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Best Of Both Worlds

Hybrid RAG: Truth Grounding + Adding Detail

Use Graph Retrieval for factual accuracy and Vector Embeddings for richness and context.

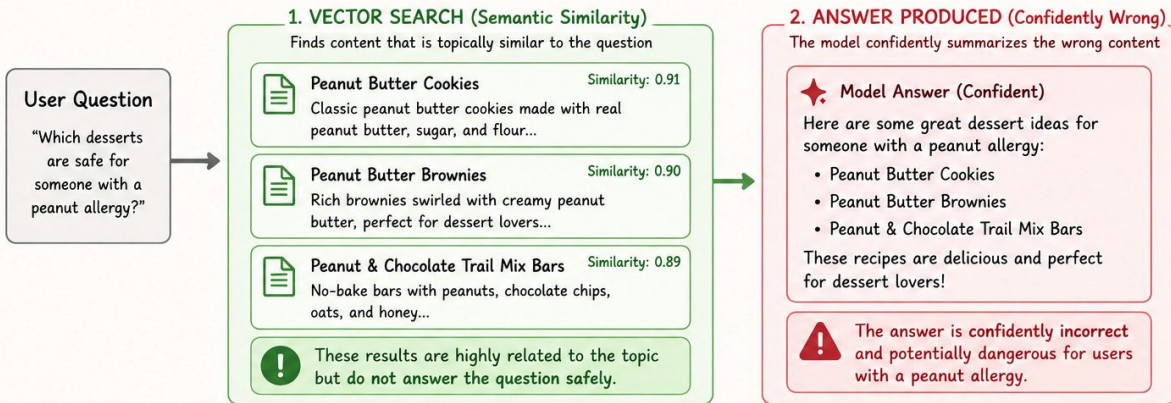


Truth Grounding + Rich Details = Better Answers

Food Allergies... Take Two!

Confidently Incorrect Answers

Semantic similarity finds related content, not the right answer.



WHY VECTOR-ONLY SEARCH FAILS



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Vector search finds what is *about* the topic, not what is *safe* or *correct*.



Use Hybrid RAG: Use keyword/structured filters on allergens and ingredients to find safe recipes, then use vector search to add helpful context like taste, substitutes, and cooking tips.

Demo: Hybrid Retrieval

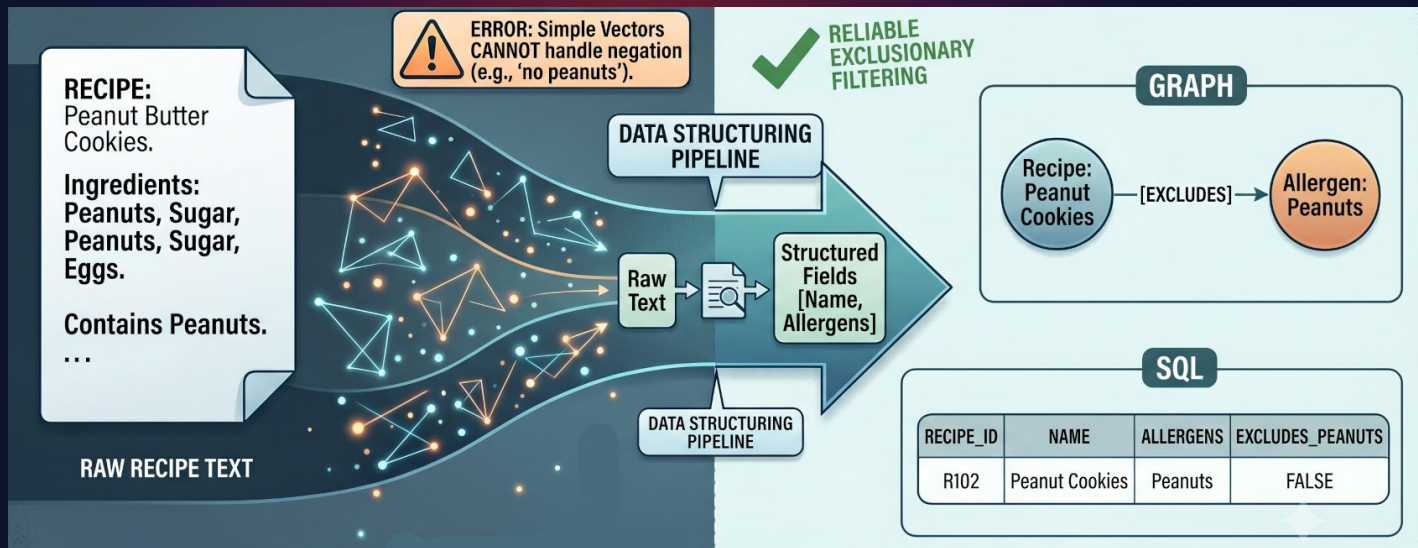
<https://youtu.be/Ah1v2YruLJQ>

Some Alternate

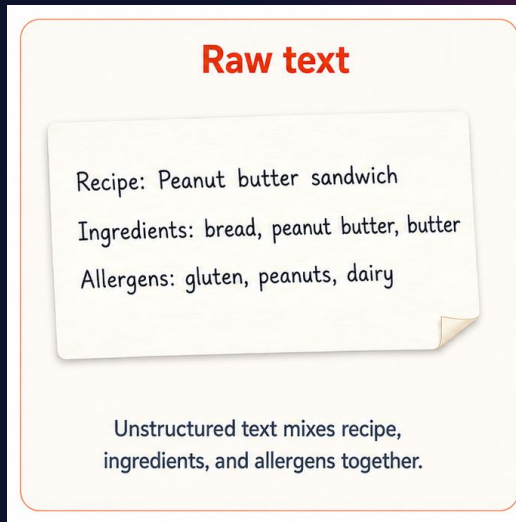
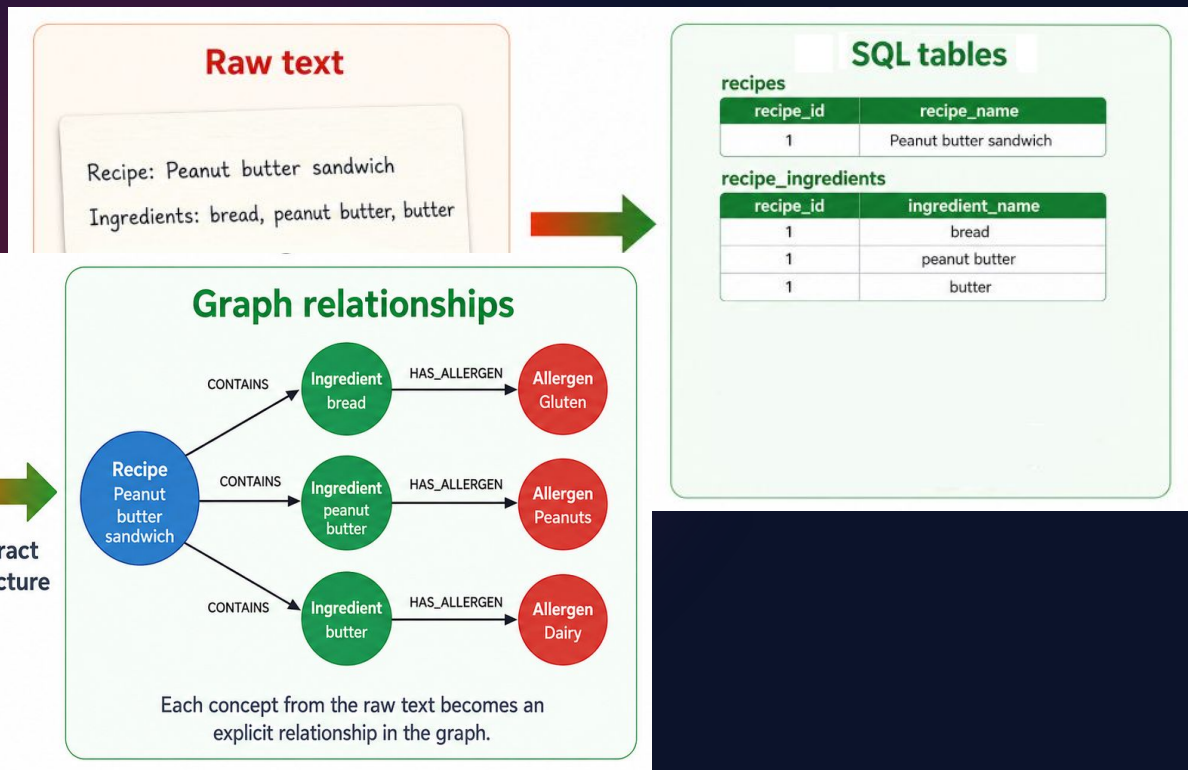
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Gemini

Options

- Unstructured → Filtering via Tags (aka Key/Value)
 - OpenSearch (sparse search), Milvus (Scalar Filtering), Quadrant (must_not)
- OpenSearch: Inverted BM25 index
- Very CPU Intensive, Doesn't Scale



Again... Consider Alternatives!



If You Structure A Lot of the Data

Raw text

Recipe: Peanut butter sandwich
Ingredients: bread, peanut butter, butter

Unstructured text mixes recipe, ingredients, and allergens together.



Extract
structure

SQL tables

recipes

recipe_id	recipe_name
1	Peanut butter sandwich

recipe_ingredients

recipe_id	ingredient_name
1	bread
1	peanut butter
1	butter

Why Not Structure All Of It!

Raw text

Recipe: Peanut butter sandwich
Ingredients: bread, peanut butter, butter
Allergens: gluten, peanuts, dairy

Unstructured text mixes recipe, ingredients, and allergens together.



Extract
structure

SQL tables

recipes

recipe_id	recipe_name
1	Peanut butter sandwich

recipe_ingredients

recipe_id	ingredient_name
1	bread
1	peanut butter
1	butter

recipe_allergens

recipe_id	allergen_name
1	Gluten
1	Peanuts
1	Dairy

Every concept from the raw text is separated into queryable columns and rows.

Recap: Pros & Cons

Knowledge Graphs

- Great at Factual Answer
 - Ideal for Known Answers
- Not So Great With Textual Nuance

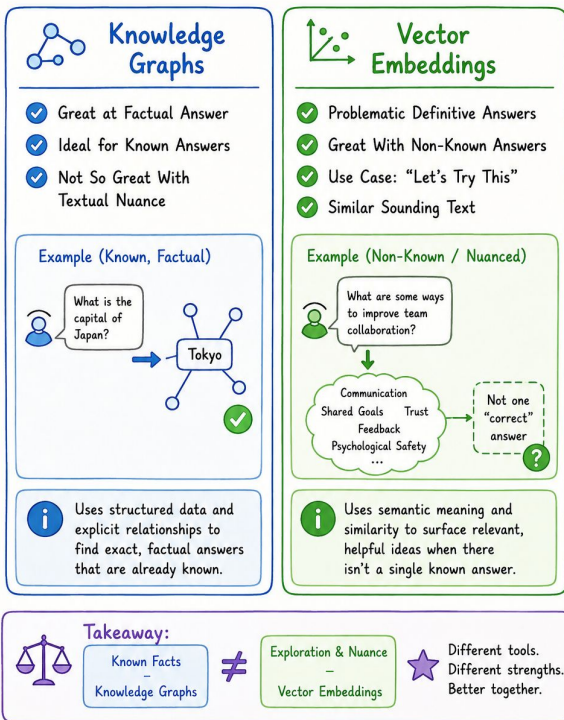
Vector Embeddings

- Problematic Definitive Answers
- Great With "Uncertain" Answers
 - Use Case: "Let's Try This"
- Similar Sounding Text

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Different Questions, Different Strengths

Some answers are known. Others aren't.
Use the right tool for the right question.



Resources

Resources

All Materials (Slides, Code, Instructions, etc):

github.com/davidvonthenen/2026-toronto-machine-learning-summit

NVIDIA [Hybrid RAG Research Paper](#)

Hybrid RAG Implementations:

- [Graph \(Neo4j\) + Vector \(OpenSearch\)](#)
- [BM25 + Vector \(OpenSearch for both\)](#)

YouTube: [Variations In RAG Implementations](#)



Thank You!



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